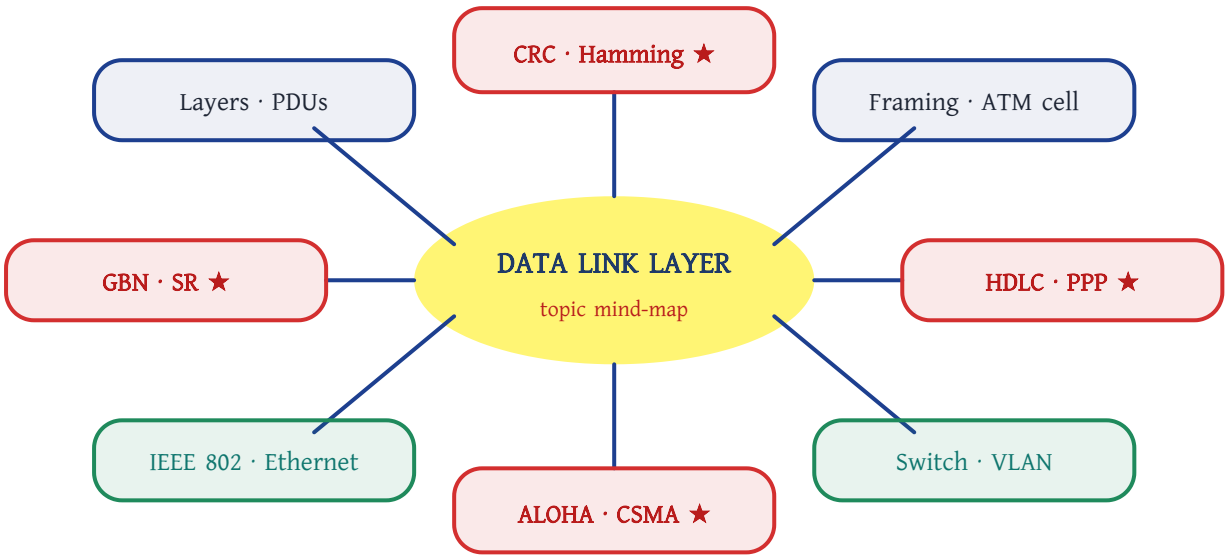
Protocol Hierarchies, Data Link & LANs

Layers · framing · CRC · Hamming · ARQ · HDLC/PPP · ATM · ALOHA/CSMA · Ethernet · switches: exam-ready in 12 sheets



↑ the whole topic on one page: red = high-yield, green = LAN practice

Syllabus coverage

"Computer Networking and Security: **Protocol hierarchies**; **Data link control: HDLC**; DLL in Internet; DLL of ATM; **LAN Protocols: Standards IEEE 802.***; Hubs, Bridges, and Switches, FDDI, Fast Ethernet; ...". Framing, error and flow control and medium access are added because every DLL question rests on them. Routing, IP, TCP, security and the full OSI-vs-TCP/IP comparison are in the other lessons of this chapter.

High-yield list

① PDU name per layer
② bit stuffing, flag 7E
③ CRC & Hamming bits
④ GBN 2^m-1, SR 2^{m-1}
⑤ ALOHA 18.4% / 36.8%
⑥ Ethernet 64–1518 B
⑦ hub L1 · switch L2

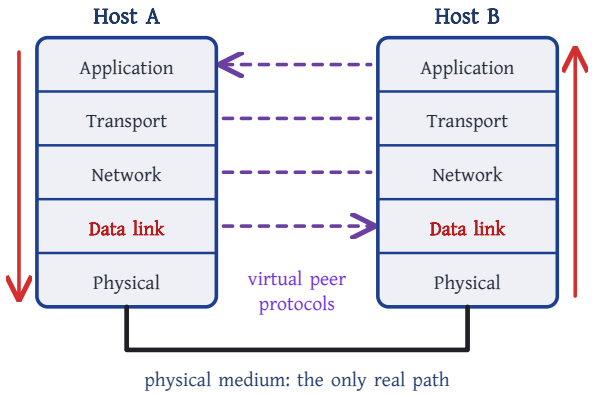
Legend: **yellow** = key term · **red circle** = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

Contents

✓ Protocol hierarchies, DLL services ... p. 2
✓ Framing, parity, checksum ..... p. 3
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1 Protocol Hierarchies: Why Layers?

Definition: A network is a **stack of layers**; each offers a **service** to the layer above and hides how. Layer n talks to layer n on the other machine (its **peer**) by the **layer-n protocol**. Architecture = layers + protocols; **protocol stack** = one protocol per layer.



- **Protocol**: rules between **peers** (horizontal)
- **Service**: what a layer gives the layer above (vertical)
- **Interface**: boundary between adjacent layers
- **SAP** (service access point): port, IP, MAC
- **Encapsulation**: each layer adds its header going down

Layer (OSI no.)	PDU	Address	Device	Example protocols
Application (7)	message	name / URL	gateway	HTTP, SMTP, DNS
Transport (4)	segment	port (16 bit)	(end hosts)	TCP, UDP
Network (3)	packet	IP (32/128 bit)	router	IP, ICMP
Data link (2)	**frame**	**MAC (48 bit)**	bridge, switch	Ethernet, HDLC, PPP
Physical (1)	bits	none	hub, repeater	RS-232, 10Base-T

Connection-oriented (set up → use → release: TCP, ATM) vs **connectionless** (full address in every unit: IP, UDP, Ethernet). **OSI primitives: Request → Indication → Response → Confirm. Design issues:** addressing, error and flow control, multiplexing, routing, fragmentation.

OSI 7 vs TCP/IP 4: full comparison in the OSI Model lesson

2 The Data Link Layer: Jobs & Services

- **Framing**: cut the bit stream into frames
- **Physical addressing**: MAC in header
- **Error control**: detect (CRC), retransmit
- **Flow control**: don't swamp the receiver
- **Access control**: who uses a shared link

IEEE sublayers:

LLC (802.2): flow/error control, which network protocol

MAC (802.3, 802.5, 802.11 ...): access + framing on one medium

Services: **unacknowledged connectionless** (Ethernet) · **acknowledged connectionless** (Wi-Fi) · **acknowledged connection-oriented** (HDLC). DLL delivery is **hop-to-hop**; transport is end-to-end.

Exam trap: The data link layer adds a **trailer** (the FCS) as well as a header; the layers above add headers only.

3 Framing: Where Does a Frame Start?

Method	How	Weakness / user
Byte count	header field gives the frame length	one bad count → **sync lost**
Flag + **byte stuffing**	FLAG at both ends; ESC before FLAG/ESC in data	byte-oriented: PPP, BISYNC
Flag + **bit stuffing**	flag **01111110**; 0 after five 1s	bit-oriented: HDLC, SDLC
Coding violations	symbols that never occur in data	4B/5B (Fast Ethernet, FDDI)
Fixed size	every frame the same length	ATM 53-byte cells



bit stuffing: the flag pattern can never appear inside the frame

Memory trick: "**Five ones, then a zero.**" Receiver, after 11111: next 0 → delete it; 01111110 → flag; seven 1s → abort.

Byte stuffing ✍️: A FLAG B → A **ESC** FLAG B · A ESC B → A **ESC** ESC B · A ESC FLAG B → A ESC ESC ESC FLAG B

Exam trap: The 0 is stuffed after five 1s **even if the next data bit is already 0**; the receiver always removes it. Worst case: 1 extra bit per 5 data bits.

4 Errors, Parity & Checksum

Burst error length = first to last bad bit (1 μs of noise at 10 Mbps hits 10 bits). **Detection** + resend = ARQ (LANs, TCP); **correction** = FEC (satellite, broadcast, ECC memory).

Method	Detects	Misses	Used in
1-D parity	any **odd** number of bit errors	any **even** number	serial ASCII
2-D parity	all 1-, 2-, 3-bit errors; corrects 1	some 4-bit	block codes
Checksum	16-bit 1's-complement sum	swapped words	IP, TCP, UDP
CRC	all bursts ≤ r bits, most others	very few	Ethernet, HDLC, PPP

Checksum ✍️ (4-bit words): Data 7, 11, 12, 0, 6 → sum = 36 = 10 0100_2 → wrap the carry: 0100 + 10 = 0110 (6) → checksum = complement = **1001 (9)**. Receiver: 36 + 9 = 45 = 10 1101 → 1101 + 10 = 1111 → complement 0000 → accept.

5 CRC (Cyclic Redundancy Check)

Method: Generator G(x) of degree **r** (r + 1 bits): append **r zeros**, divide by G in **modulo-2** (XOR); the r-bit remainder is the CRC; send data + CRC. Receiver divides by G: **remainder 0 → accept**.

Worked example ✍️: Data 100100, G = 1101 ($x^3 + x^2 + 1$, r = 3)

1. Dividend 100100**000**
2. 1001⊕1101 = 0100 → 1000⊕1101 = 0101
3. 1010⊕1101 = 0111 → 1110⊕1101 = 0011
4. 0110 (leading 0: ⊕0000) → 1100⊕1101 = **001** → send **100100 001**

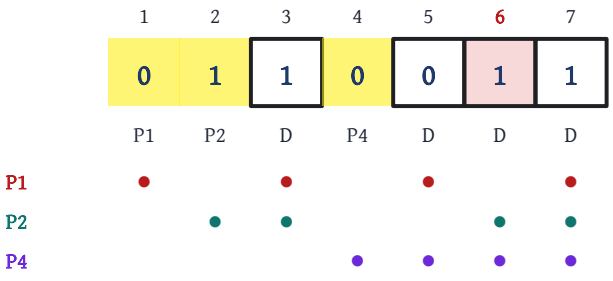
CRC catches:
all single-bit errors · all odd-count errors if (x + 1) divides G · **all bursts ≤ r** · burst r + 1: 1 − (½)^(r−1) · longer: 1 − (½)^r
CRC-8 ATM HEC · **CRC-16** HDLC, PPP · **CRC-32** Ethernet, Wi-Fi

Exam trap: Append **degree** zeros (3 for 1101), not the number of generator bits (4). Modulo-2 subtraction = XOR: no borrows.

6 Hamming Distance & Hamming Code

Hamming distance: d(x, y) = positions where two codewords differ = 1s in x ⊕ y; d(10101, 11110) = 01011 → **3**. d_{min} = smallest distance between any two valid codewords.

Rules: detect s: $d_{min} \geq s + 1$ · correct t: $d_{min} \geq 2t + 1$ · check bits: $2^r \geq m + r + 1$ → m = 4: 3 (7, 4) · 7: 4 (11, 7) · 8: 4 (12, 8) · 16: 5 · 32: 6



Hamming (7, 4) for data 1011: who checks whom

Worked example ✍️: Data 1011 into positions 3, 5, 6, 7 (even parity):

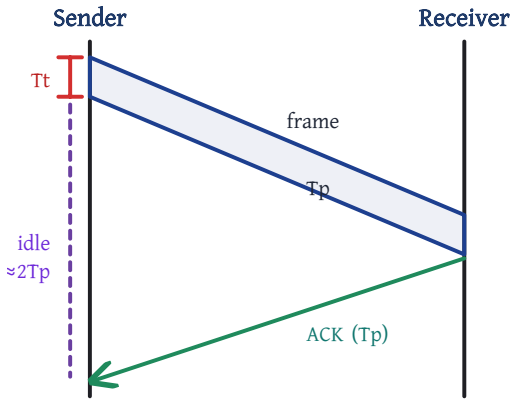
1. P1 = 3⊕5⊕7 = 1⊕0⊕1 = 0
2. P2 = 3⊕6⊕7 = 1⊕1⊕1 = 1
3. P4 = 5⊕6⊕7 = 0⊕1⊕1 = 0 → **0110011**
4. Bit 6 flips: 0110001 → checks C4 C2 C1 = 1 1 0 = **6** → flip bit 6

Memory trick: Parity bits sit at **powers of 2**; position k is checked by the ones that sum to k: 6 = 4 + 2.

Exam trap: Hamming (7, 4), $d_{min} = 3$: corrects 1 **or** detects 2. An extra parity bit gives SEC-DED (ECC RAM).

7 Flow Control: Stop-and-Wait

Definition: **Flow control** stops a fast sender flooding a slow receiver. **Error control** = **ARQ** (automatic repeat request): detect, ACK/NAK, timeout, resend. **Stop-and-wait**: one frame, then wait for its ACK; seq. numbers 0, 1 catch duplicates.



the line sits idle for 2Tp per frame

Formulas:

Tt = L / B (frame bits ÷ bandwidth)

Tp = d / v (v ≈ 2×10^8 m/s in cable)

a = Tp / Tt

Stop-and-wait: **η = 1 / (1 + 2a)**

Sliding window: η = W / (1 + 2a), max 1

100% busy needs **W ≥ 1 + 2a**

Throughput = η × B · BDP = B × RTT

Worked example ✍️: B = 1 Mbps, L = 1000 bits, Tp = 20 ms. Tt = 1 ms → a = 20 → stop-and-wait η = 1/41 = **2.4%** (24 kbps). Full use needs W ≥ 41 → Go-Back-N: $2^m - 1 \geq 41$ → **m = 6**; Selective Repeat: $2^{m-1} \geq 41$ → **m = 7**.

8 Sliding Window: Go-Back-N vs Selective Repeat

Point	Stop-and-Wait	Go-Back-N	Selective Repeat
Sender window	1	**$2^m - 1$**	**2^{m-1}**
Receiver window	1	1	2^{m-1}
Frame lost	resend it	resend it **and all after it**	resend **only** that frame
ACK style	one per frame	cumulative	individual (+ NAK)
Out-of-order frames	n/a	discarded	buffered

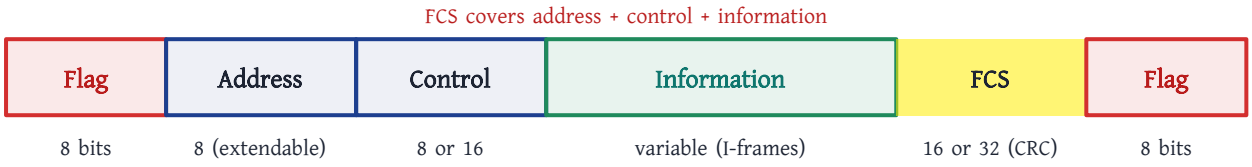
m = 3 → seq. 0–7: GBN window **7**, SR window **4**. **Piggybacking**: the ACK rides in a reverse data frame (HDLC N(R)). GBN wastes bandwidth on errors; SR needs most buffers.

Memory trick: GBN: "**all but one**" ($2^m - 1$). SR: "**half**" (2^{m-1}).

Exam trap: SR window over half → an old and a new frame share a **number**.

9 HDLC (High-Level Data Link Control)

Definition: ISO **bit-oriented** protocol (from IBM's SDLC), point-to-point or multipoint, half/full duplex: **bit stuffing**, flag **01111110 (7E)**, sliding-window ARQ. Family: **LAPB** (X.25), **LAPD** (ISDN), LAPM (modems); basis of PPP and 802.2 LLC framing.



HDLC frame: one flag may close one frame and open the next

Stations: primary sends commands, **secondary** sends responses, **combined** does both. **Configurations: unbalanced** = 1 primary + n secondaries; **balanced** = 2 combined stations (point-to-point).

Transfer mode	Config.	Who may start sending?
NRM Normal Response Mode	unbalanced	secondary only when **polled** (multipoint)
ARM Asynchronous Response Mode	unbalanced	secondary without permission (rare)
ABM Asynchronous Balanced Mode	balanced	either station, any time (LAPB; most common)

Control field (8-bit, modulo 8)

Frame	Bits 1 → 8	Job
I-frame	**0** · N(S) 3 · P/F · N(R) 3	user data + piggybacked ACK
S-frame	**10** · code 2 · P/F · N(R) 3	ACK / NAK, flow control (no data)
U-frame	**11** · M 2 · P/F · M 3	link management: SNRM, SABM, DISC, UA, FRMR

S code	Name	Meaning
00	RR	ACK, ready
10	RNR	ACK, but stop
01	REJ	NAK: Go-Back-N
11	SREJ	NAK: Selective

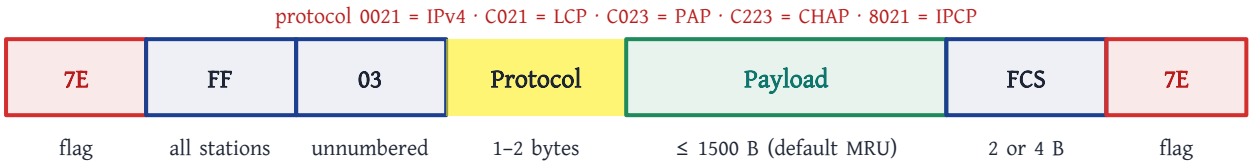
3-bit N(S)/N(R) → window ≤ **7**; extended 16-bit control (SABME) → modulo 128, window ≤ 127. **P/F**: Poll in a command, Final in a response.

Memory trick: I-S-U start with **0, 10, 11.** RR/RNR = "Ready / Not Ready", REJ = reject all, SREJ = reject one.

Exam trap ✍️: N(R) = the **next frame expected**, not the last received: an I-frame with N(S) = 3, N(R) = 5 is frame 3 and acknowledges frames up to 4. HDLC is bit-oriented; PPP looks similar but is **byte-oriented**.

10 DLL in the Internet: SLIP & PPP

Point-to-point links (dial-up, leased lines, DSL as PPPoE, router-to-router over SONET) use **PPP** (RFC 1661), the successor of SLIP. On LANs the Internet runs over Ethernet and Wi-Fi.



PPP frame: HDLC-like, but byte-oriented

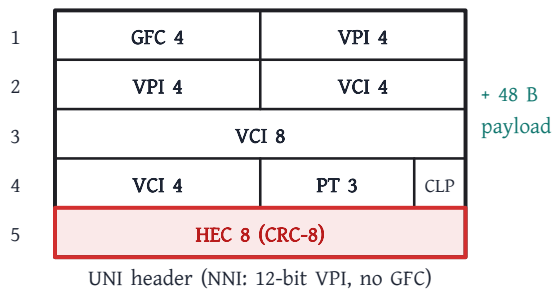
Point	SLIP (RFC 1055)	PPP (RFC 1661)
Framing	END byte C0, escape DB	flag 7E, escape **7D**, byte ⊕ 20
Error detection	none	FCS (CRC-16 or CRC-32)
Network protocols	IP only	many (protocol field, NCPs)
IP address	configured by hand	negotiated (IPCP)
Authentication	none	PAP or CHAP

PPP phases: Dead → Establish (**LCP**: link options) → Authenticate (optional) → Network (**NCP**, e.g. IPCP) → Open → Terminate → Dead. **PAP**: 2-way, password in clear text, once. **CHAP**: 3-way challenge–response (MD5 hash of challenge + secret), secret never sent, repeated during the session. Stuffing: 7E → 7D 5E, 7D → 7D 5D.

> **Exam trap:** PPP **detects** errors (FCS) but in its default mode has no sequence numbers, ACKs or flow control: a bad frame is just dropped and upper layers recover.

11 Data Link Layer of ATM

> **Idea:** ATM switches fixed **53-byte cells** = 5-byte header + 48-byte payload over virtual circuits (VPI/VCI instead of MAC addresses). The data-link-type work is done by the **TC (Transmission Convergence) sublayer** under the ATM layer.



UNI header (NNI: 12-bit VPI, no GFC)

- **HEC**: CRC-8 ($x^8 + x^2 + x + 1$) over the first **4 header bytes** only; corrects 1-bit, detects multi-bit header errors
- **Cell delineation**: find cell edges by checking HEC (hunt → presync → sync)
- Cell-rate decoupling (idle cells), fit cells into SONET/SDH frames
- **CLP = 1**: drop this cell first under congestion

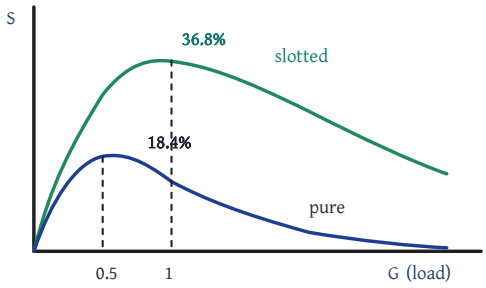
Memory trick: "**5 + 48 = 53**". 48 was a compromise between 32 and 64 bytes. Cell tax = 5/53 ≈ **9.4%**.

Elsewhere: AAL → Transport lesson; ATM network layer (VPI/VCI switching) → IP lesson.

12 Medium Access Control: Who Talks Next?

Random access: ALOHA, CSMA, CSMA/CD, CSMA/CA · **Controlled access:** reservation, polling, **token passing** · **Channelization:** FDMA, TDMA, CDMA.

	Pure ALOHA	Slotted ALOHA
Send	any time	slot start only
Vulnerable	$2 \times T_{fr}$	T_{fr}
Throughput S	$G \cdot e^{-2G}$	$G \cdot e^{-G}$
Max S	**18.4%**	**36.8%**
at load	G = 0.5	G = 1



Worked example ✍️: 200-bit frames on a 200 kbps channel → T_{fr} = 1 ms. Load 1000 frames/s → G = 1. Pure: $S = e^{-2} = 0.135$ → **135 frames/s** succeed. Slotted: $S = e^{-1} = 0.368$ → **368 frames/s**.

CSMA type	Channel idle	Channel busy	Note
1-persistent	send at once	keep sensing, send when idle	Ethernet; most collisions
Non-persistent	send at once	wait random time, sense again	fewer collisions, more delay
p-persistent	send with probability p	wait for next slot	slotted channels

CSMA/CD (classic half-duplex Ethernet)

Listen before and **while** sending; on a collision send a **jam** signal, abort, back off. To detect a collision the sender must still be sending when it returns: $T_{fr} \geq 2 \times T_p \rightarrow L_{min} = 2 \times T_p \times B$.

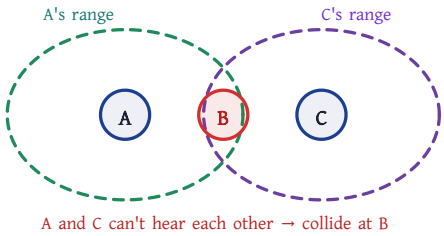
Example ✍️: 10 Mbps, worst round trip 51.2 µs → $L_{min} = 10^7 \times 51.2 \times 10^{-6} = 512$ **bits = 64 bytes**, the Ethernet minimum frame.

Binary exponential backoff: after the n-th collision wait K × 512 bit-times, K random in 0 ... 2^min(n,10) - 1; give up after **16** tries.

Exam trap: Full-duplex switched Ethernet has **no collisions**, so CSMA/CD is switched off. Carrier sensing cannot stop all collisions because of propagation delay.

CSMA/CA (wireless): A radio cannot hear a collision while it transmits (its own signal swamps the receiver), and some stations cannot hear each other. So Wi-Fi **avoids** collisions: interframe spaces, a random contention window and a **positive ACK** for every frame.

13 IEEE 802.11 (Wi-Fi)



hidden-station problem → cured by RTS/CTS

- **RTS/CTS**: sender asks (RTS), receiver grants (CTS); all who hear CTS keep quiet
- **NAV**: virtual carrier sense, the "busy until" timer
- IFS order: **SIFS < PIFS < DIFS**; ACK/CTS wait only SIFS, so they win
- **DCF** (contention, mandatory) · **PCF** (AP polls, optional)
- Frame has up to **4 MAC addresses** (via the AP)

802.11b 2.4 GHz, 11 Mbps, DSSS · **a** 5 GHz, 54 Mbps, OFDM · **g** 2.4 GHz, 54 Mbps, OFDM · **n** (Wi-Fi 4) 2.4/5 GHz, 600 Mbps, MIMO · **ac** (Wi-Fi 5) 5 GHz, ≈ 6.9 Gbps, MU-MIMO · **ax** (Wi-Fi 6/6E) up to 6 GHz, ≈ 9.6 Gbps, OFDMA. BSS = AP + stations; ESS = BSSs on a distribution system; IBSS = ad hoc. Security: WEP (broken) → WPA → **WPA2** (802.11i, AES-CCMP) → WPA3.

Memory trick: Wired = **D**etect (CD), wireless = **A**void (CA). "**S**hort waits win": SIFS is shortest.

14 Token Passing: Token Ring & FDDI

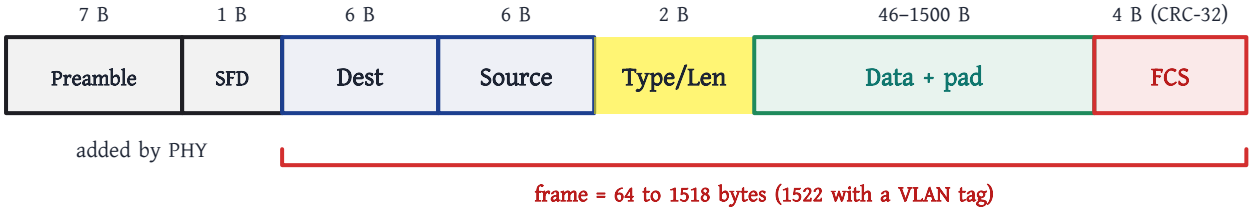
Token Ring (IEEE 802.5): 4 or 16 Mbps; a 3-byte **token** circulates and only its holder sends; differential Manchester; star-wired ring through an MAU; **no collisions, deterministic delay**. Token Bus = 802.4.

FDDI (ANSI X3T9.5): **100 Mbps** over fibre; **dual counter-rotating rings** (on a break the ring wraps); timed token; 4B/5B + NRZ-I; ring ≤ 100 km, ≈ 500 stations. **Trap:** ANSI, not IEEE 802.

15 The IEEE 802 Family

Standard	Covers	Standard	Covers
802.1D	bridges, Spanning Tree	**802.3**	Ethernet (CSMA/CD)
802.1Q · 802.1X	VLAN tag · port access control	802.4 · 802.5	Token Bus · Token Ring
802.2	LLC	802.6	DQDB (MAN)
802.3u	Fast Ethernet 100 Mbps	**802.11**	wireless LAN (CSMA/CA)
802.3z / ab	Gigabit (fibre / UTP)	802.15	WPAN: .1 Bluetooth, .4 (ZigBee)
802.3ae · af	10 GbE · Power over Ethernet	802.16	WiMAX (wireless MAN)

16 Ethernet Frame & MAC Address



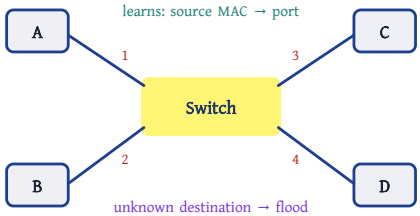
- Preamble 10101010 × 7, SFD **10101011** (not counted); data < 46 B is **padded**; gap 96 bit-times. Type ≥ 0x0600 → **EtherType** (0800 IPv4, 0806 ARP, 86DD IPv6); ≤ 1500 → length.
- **MAC address 48 bits**: first 24 = **OUI** (vendor); LSB of byte 1: 0 unicast, 1 multicast; broadcast **FF-FF-FF-FF-FF-FF**.

Media: 10Base5 thick coax 500 m · 10Base2 thin coax 185 m · 10Base-T UTP 100 m · 10Base-F 2 km (Manchester) · Fast Ethernet 802.3u: **100Base-TX** Cat 5, 100 m, 4B/5B + MLT-3 · 100Base-FX fibre 2 km · 100Base-T4 Cat 3, 8B/6T.

> **Exam trap:** Minimum frame **64 B**, minimum data **46 B**; max frame 1518 B, MTU 1500 B. Fast Ethernet keeps the 64-byte minimum.

17 Hubs, Bridges, Switches & VLANs

Device	Layer	Forwards by	Collision dom.	Broadcast dom.
Repeater / **hub**	1	nothing: repeats to all ports	1 for all ports	1
Bridge / **switch**	2	MAC address	1 per port	1 (per VLAN)
Router / L3 switch	3	IP address	1 per port	**1 per port**



Transparent bridge (802.1D): **backward learning** of each **source** MAC → port; then **forward** to the known port, **filter** if same port, **flood** if unknown/broadcast. Switch modes: **store-and-forward** (CRC checked) · **cut-through** (after dest. MAC) · **fragment-free** (first 64 B).

> **STP (802.1D):** Loops cause **broadcast storms**; STP elects a **root bridge** (lowest priority + MAC) and blocks ports → one loop-free tree. RSTP = 802.1w.

> **VLAN (802.1Q):** A software **broadcast domain**. 4-byte tag: TPID 8100, PCP 3, DEI 1, VID 12 bits → **4094**; trunks carry tags.

> **Worked example ✍️:** Router → switch (4 PCs) and → hub (3 PCs). Collision domains = 5 switch ports + 1 hub = **6**; broadcast domains = **2**.

Night-Before-the-Exam Sheet

Fact sheet:

PDU: bits → frame → packet → segment

DLL adds header + trailer (FCS)

Bit stuffing: 0 after five 1s · flag 7E

PPP escape 7D, byte ⊕ 20 · SLIP C0

Parity: catches odd-count errors only

CRC: degree r → r zeros, bursts ≤ r

Detect s: $d_{min} \geq s + 1$

Correct t: $d_{min} \geq 2t + 1$

Hamming: $2^r \geq m + r + 1$

m = 4 → 3 · 7 → 4 · 8 → 4 · 16 → 5

a = Tp / Tt · S&W $\eta = 1/(1 + 2a)$

window $\eta = W/(1 + 2a)$; $W \geq 1 + 2a$

GBN $W \leq 2^m - 1$ · SR $W \leq 2^{m-1}$

HDLC NRM · ARM · ABM; I/S/U

S-frames RR 00 · REJ 01 · RNR 10 · SREJ 11

ATM 53 = 5 + 48; HEC = CRC-8

Pure ALOHA 18.4% at G = 0.5

Slotted ALOHA 36.8% at G = 1

CSMA/CD: $L_{min} = 2 \cdot Tp \cdot B$

Backoff 0 … 2^min(n,10) − 1; quit at 16

Ethernet 64–1518 B, data 46–1500

MAC 48 bit, OUI 24 · bcast FF…FF

SIFS < PIFS < DIFS · RTS/CTS

FDDI 100 Mbps dual ring (ANSI)

802.3 · 802.5 · 802.11 · 802.1Q

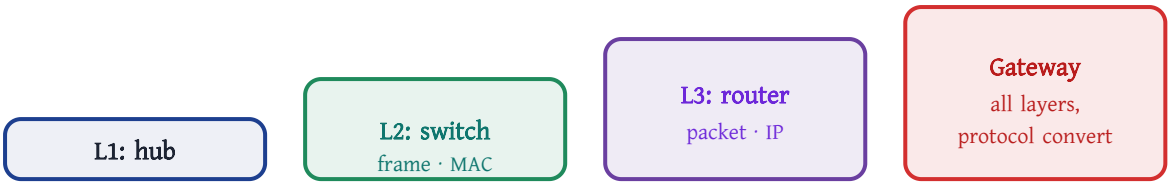
VLAN tag 4 B, VID 12 bit, 4094

Top 12 one-liners for MCQs

1. MAC address = 48 bits
2. Ethernet = IEEE 802.3, Wi-Fi = 802.11
3. Token Ring = 802.5, Token Bus = 802.4
4. HDLC is bit-oriented; PPP byte-oriented
5. 3-bit seq: GBN window 7, SR window 4
6. Wi-Fi uses CSMA/CA, not CSMA/CD
7. Hub L1 · bridge, switch L2 · router L3
8. Router separates broadcast domains
9. ATM HEC protects the header only
10. CHAP: 3-way, secret never sent
11. STP = 802.1D; VLAN = 802.1Q
12. Fast Ethernet 802.3u, 100Base-TX

Top traps: CRC zeros = degree, not generator length · N(R) = next frame expected · Hamming (7, 4) corrects 1 **or** detects 2 · stuff the 0 even if the next bit is 0 · full-duplex Ethernet has no CSMA/CD · minimum frame 64 B vs minimum data 46 B · FDDI is ANSI, not IEEE

All tricks together: Five ones, then a zero · parity bits at powers of 2 · GBN all but one, SR half · I-S-U = 0, 10, 11 · 5 + 48 = 53 · wired Detects, wireless Avoids · short waits win (SIFS) · hub hears all, switch splits collisions, router splits broadcasts



the device staircase: each step up reads one more header

✓ Practice MCQs (exam pattern, 10 marks)

1. Go-Back-N with 4-bit sequence numbers: maximum sender window?
(a) 16 (b) 8
(c) 15 (d) 7

2. Maximum throughput of pure ALOHA:
(a) 18.4% (b) 36.8%
(c) 50% (d) 100%

3. HDLC bit stuffing inserts a 0 after:
(a) four 1s (b) six 1s
(c) eight 1s (d) five consecutive 1s

4. The HEC field of an ATM cell protects:
(a) the 48-byte payload (b) the cell header only
(c) the whole 53-byte cell (d) the AAL trailer

5. Minimum Ethernet frame size:
(a) 64 bytes (b) 46 bytes
(c) 512 bytes (d) 1518 bytes

6. Which device separates broadcast domains?
(a) hub (b) switch
(c) router (d) bridge

7. Hamming check bits needed for 7 data bits:
(a) 3 (b) 5
(c) 7 (d) 4

8. Medium access method of IEEE 802.11:
(a) CSMA/CD (b) CSMA/CA
(c) token passing (d) pure ALOHA

9. IEEE 802.1Q defines:
(a) Spanning Tree (b) LLC
(c) VLAN tagging (d) Token Ring

10. HDLC frames used for link set-up and disconnect:
(a) U-frames (b) I-frames
(c) S-frames (d) P-frames

Answer key → 1-c · 2-a · 3-d · 4-b · 5-a · 6-c · 7-d · 8-b · 9-c · 10-a

Written question: A 1 Mbps satellite link has a one-way propagation delay of 270 ms and carries 1000-bit frames. Find (i) the link utilization with stop-and-wait, (ii) the window size for full utilization, (iii) the minimum sequence-number bits for Go-Back-N and for Selective Repeat. 8 marks

1. $T_t = L/B = 1000 / 10^6 = 1$ ms; $a = T_p/T_t = 270$ 1
2. Stop-and-wait $\eta = 1/(1 + 2a) = 1/541 \approx$ **0.18%** (throughput ≈ 1.85 kbps) 2
3. Full use: $W \geq 1 + 2a =$ **541 frames** (bandwidth × RTT = 541 000 bits) 2
4. GBN: $2^m - 1 \geq 541 \rightarrow$ **m = 10** (window 1023). SR: $2^{m-1} \geq 541 \rightarrow$ **m = 11** (window 1024) 3

Before the exam ✓

- ✓ Can name PDU, address and device for each layer
- ✓ Can bit-stuff and byte-stuff a data stream
- ✓ Can compute a CRC and a Hamming (7, 4) code

Also practise (written):
① CRC of 1101 with G = 1011
② HDLC modes and frame types
③ CSMA/CD minimum frame, 1 Gbps, 1 km
④ hidden station and RTS/CTS

✓ Can count collision and broadcast domains